Calculus in Signal Processing: Understanding and Analyzing Signals

1. Abstract

Every signal – a voice, a heartbeat on an ECG, a Wi-Fi carrier – is a function of time. This project shows how the core tools of calculus (derivatives, integrals and infinite series) let engineers measure, decompose and clean up such signals. Each concept is linked to a real signal-processing task and verified with Python simulations.

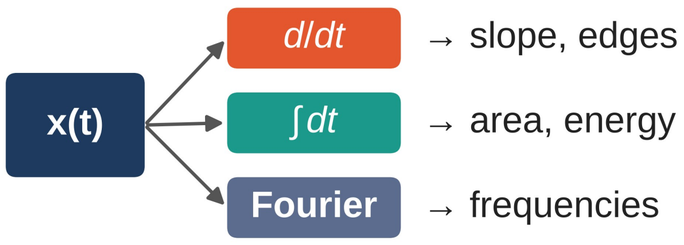
2. Introduction

Signal processing keeps asking two questions: how fast is the signal changing, and how much of it has built up?

The **derivative** dx/dt answers the first – it reveals slopes, sharp edges and sudden events.

The **integral** ∫x(t)dt answers the second – it gives area, average value and energy.

Joseph Fourier then showed that any periodic signal is a sum of sines and cosines whose weights are found by integration. Together these ideas turn a messy waveform into numbers we can analyse and design with (Fig. 1).



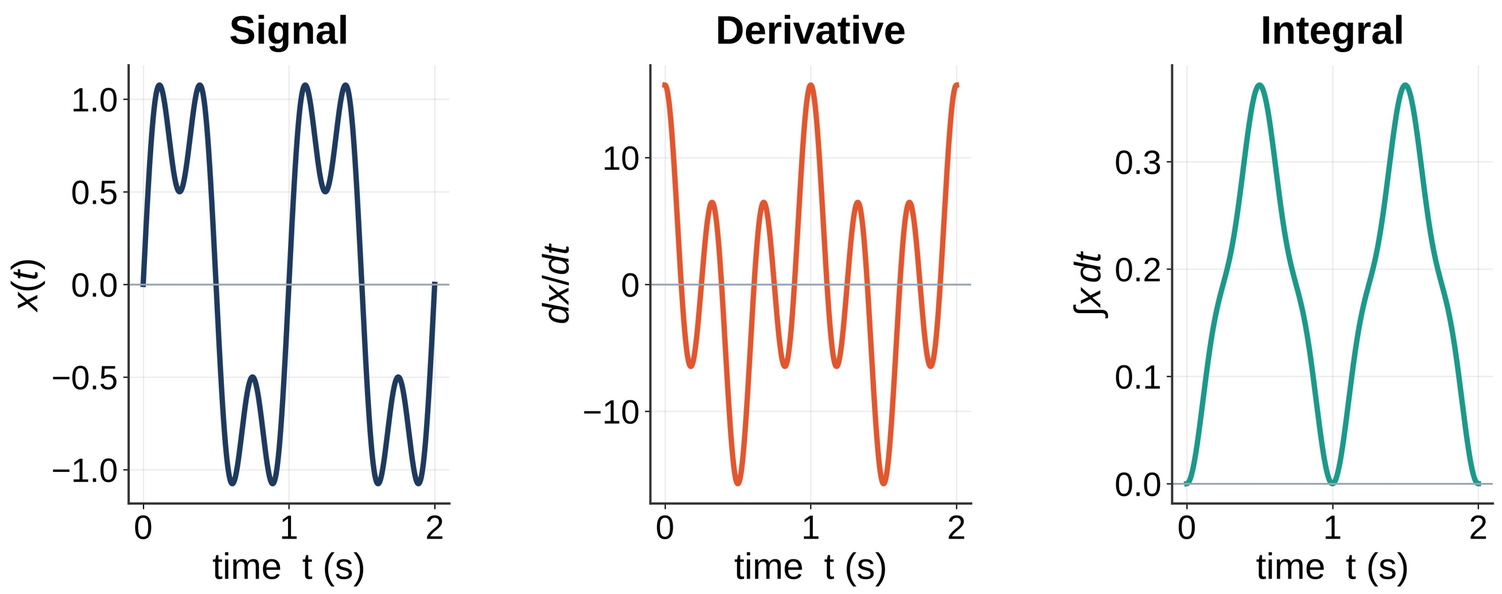


Fig. 1 : x(t) = sin 2πt + 0.5 sin 6πt with its derivative and running integral

3. Concept: Key Calculus Tools

- **Derivative:** x′(t) = lim (x(t+h) − x(t)) / h → rate of change
- **Energy:** E = ∫ |x(t)|² dt → area under |x|²
- **Fourier series:** $a_n = (2/T) \int x(t) \cos(n\omega_0 t)\, dt$
- **Fourier transform:** $X(f) = \int x(t)\, e^{-j2\pi ft}\, dt$
- **Convolution:** $y(t) = \int x(\tau)\, h(t-\tau)\, d\tau$
- **Laplace:** d/dt ↔ s, so differential equations become algebra

Hypothesis: a computer can replace these limits with finite differences and Riemann sums (Fig. 3) and still give accurate results.

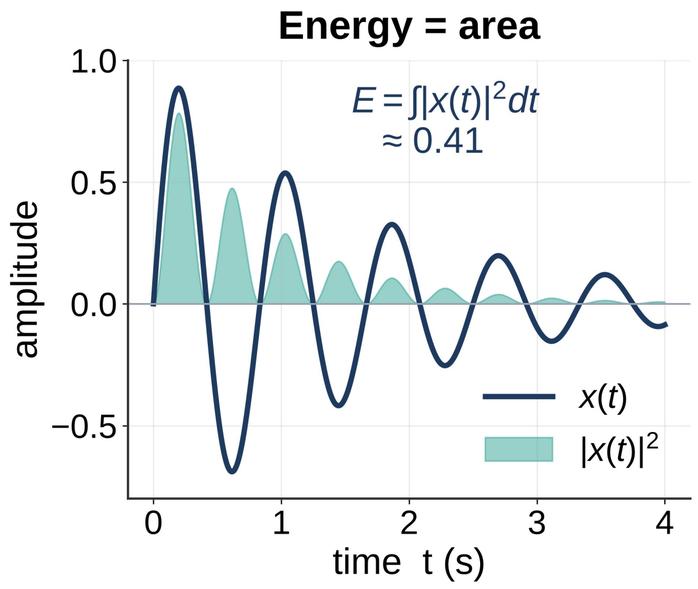


Fig. 2 : Energy is the shaded area

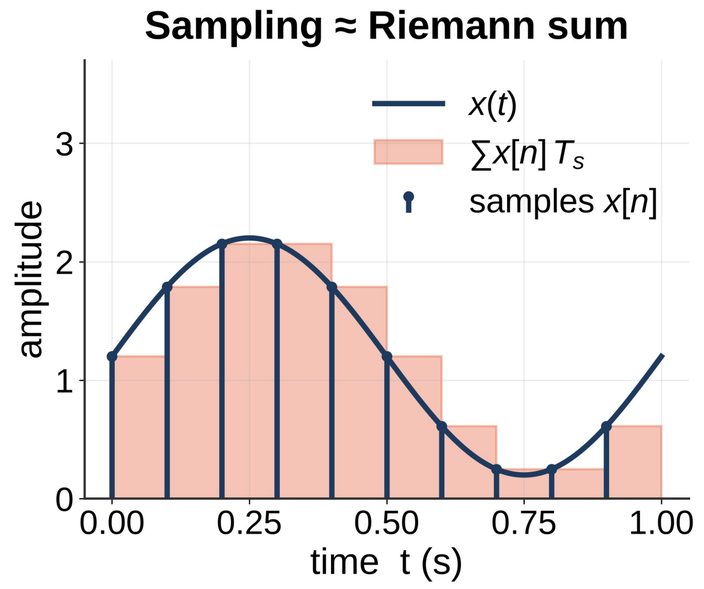


Fig. 3 : Samples approximate the integral

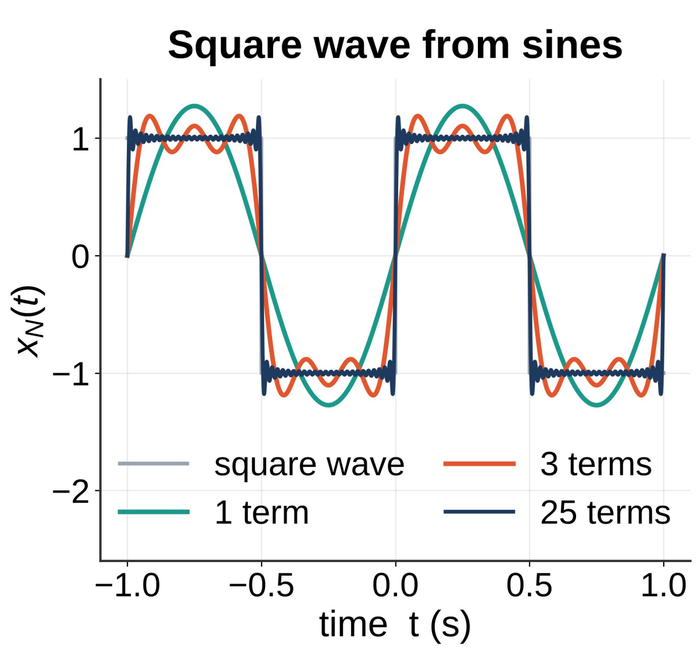


Fig. 4 : Partial Fourier sums

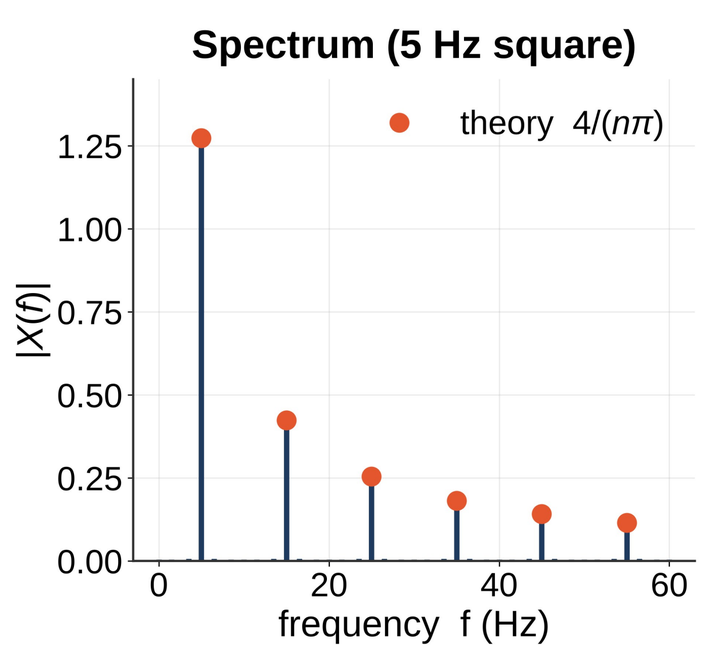


Fig. 5 : FFT spectrum vs theory

4. Method & Results

- Signals were simulated in Python (NumPy); integrals used the trapezoidal rule, derivatives used finite differences.
- Square wave: $b_n = 4/(n\pi)$ for odd n. With 25 terms the sum matches the wave except a ~9% overshoot at each jump (Gibbs effect).
- The FFT shows peaks at 5, 15, 25 Hz … with heights matching 4/(nπ).
- An RC low-pass filter, RC dy/dt + y = x (f_c = 12 Hz), removes most noise from a 3 Hz sine (Fig. 6).

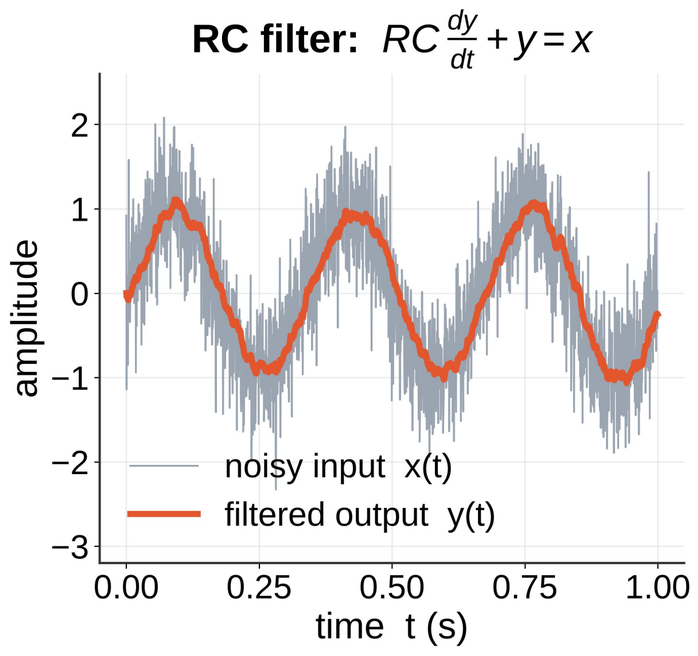


Fig. 6 : Noise removed by an RC filter

5. Conclusion

Calculus is the language of signal processing: derivatives expose change, integrals measure energy, and Fourier analysis reveals frequency. Digital systems – ECG monitors, audio equalisers, Wi-Fi – simply approximate these operations with sums and differences.

6. References

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3. S. W. Smith, The Scientist and Engineer's Guide to Digital Signal Processing, 1997.

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